

Lesson 64 Practice: Answers and Rationales

JKSSB Computer Awareness

October 10, 2026

Provenance: All 25 items are original exam-pattern questions (*[Original]* plus the *[JKSSB FAA-pattern]*, *[JKPSI-pattern]*, *[SSC CGL-pattern]*, and *[JKSSB-pattern]* extension items, including MSQs); none reproduces an official paper item.

- Q1. A.** A parasitic (file-infector) virus attaches to an executable file and runs when the host runs. B describes a worm (no host needed); C describes a Trojan (disguise, no self-replication); D describes DDoS.
- Q2. A.** DES is the Data Encryption Standard. B, C, and D are not the correct full form.
- Q3. C.** DES works on 64-bit blocks. A (32) and D (128) are other ciphers' sizes; B (56) is the effective key size, not the block size.
- Q4. D.** A signature proves authenticity (A), non-alteration (B), and integrity (C); it does not hide the body — that is encryption. D is the false statement (TRAP-15).
- Q5. A.** Encryption gives confidentiality; a signature gives authenticity and integrity. B swaps the pair; C and D use availability, which belongs to the DDoS threat, not to these two verbs.
- Q6. A.** Per-packet header checks are packet filtering. Stateful (B) adds connection state; the proxy (C) inspects application content; an anti-virus (D) disinfects files rather than filtering packets.
- Q7. B.** Tracking connection state and dropping stray packets is stateful inspection. A decides per packet without sessions; C and D are malware types, not firewalls.
- Q8. C.** The go-between inspecting application content is the application (proxy) firewall. A checks headers only; B tracks sessions; D is only a mail-client feature.
- Q9. A.** A fake bank/KYC mail harvesting OTPs is phishing. B locks files for ransom; C floods a server; D is unrelated maintenance.
- Q10. B.** Locked files plus a payment demand is ransomware. A steals credentials via trickery; C floods a service; D describes no concrete mechanism.
- Q11. B.** This option assigns each threat its correct mechanism. A, C, and D scramble the definitions (flood = DDoS, trick = phishing, lock = ransomware; a firewall or backup is not a threat).
- Q12. A.** DDoS is Distributed Denial of Service; the distributed prefix (many machines) is the tested point. B, C, and D are not expansions.
- Q13. B.** I and II are correct: DES is symmetric and uses 64-bit blocks. III is false because secrecy of the body is encryption, not signature (TRAP-15). A omits II; C and D include false III.

- Q14. B.** Stateful inspection tracks connection state plus headers. A and C swap the packet-filter and proxy roles; D assigns disinfection (anti-virus work) to a packet filter.
- Q15. A.** Cross-border applicable law (jurisdiction) is a genuine governance concern, with security, privacy, and access. B, C, and D are single-office trivia, not governance issues.
- Q16. A.** Purpose limitation restricts use to the stated purpose. Unlimited retention (B), sharing without consent (C), and default publication (D) all violate protection principles.
- Q17. A.** Unlicensed copying of software is infringement. Fair dealing (B) covers limited uses such as study; first ownership (C) is who owns first; a lawful backup (D) requires licence permission.
- Q18. B.** Meaning = owner's exclusive rights; first ownership = who owns first; infringement = what counts as violation; fair dealing = limited permitted uses such as study. A, C, and D rotate the matches.
- Q19. A.** ICANN is the Internet Corporation for Assigned Names and Numbers. B, C, and D are not the correct full form.
- Q20. A.** A multi-machine flood blocking access is DDoS, met at the network barrier. B answers phishing, not flooding; C answers a file virus with a printer driver; D treats an attack as a copyright exception.
- Q21. A.** Encryption keeps data confidential. B, C, and D are unrelated device effects.
- Q22. A.** Signatures prove who sent data and that it is unaltered. B, C, and D are non-security freebies.
- Q23. A.** The IT Act covers cybercrime. B, C, and D are invented distractors.
- Q24. A, B, C.** Shared-key use, AES, and bulk speed mark symmetric ciphers. Key pairs (D) belong to asymmetric crypto.
- Q25. A, B, C.** The encryption, decryption, and hashing roles are correct. D is false: encryption does not speed printing.
- Q26. A.** S/MIME can encrypt the message itself. IMAP retrieves mail, SMTP transfers it, and TLS/SSL on one connection hop is transport protection rather than end-to-end message encryption.
- Q27. A.** A Trojan hides a malicious purpose inside something that appears benign. Self-replication describes a worm; the other options describe defensive controls.
- Q28. A.** Encryption protects confidentiality. Availability concerns access; signatures can support non-repudiation/integrity; compression reduces size.
- Q29. A.** A digital signature supports sender authenticity and message integrity. Encryption, not signing alone, provides confidentiality.
- Q30. A, B, C.** The first three are standard roles. Antivirus reduces risk but cannot guarantee prevention of every attack.